

Installing and provisioning SQL Server

T2765

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Overview of SQL Server

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Managing database storage and configuration options

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Provisioning SQL Databases

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Facilities

- Class hours
- Building hours
- Restrooms
- Meals

About This Course

- Audience
- Course Prerequisites
- Course Objectives
- What You Can Expect

Course Outline

- **Module 1**, SQL Server components
- **Module 2**, Installing SQL server
- **Module 3**, Upgrading SQL Server
- **Module 4**, Working with databases
- **Module 5**, Performing Database Maintenance
- **Module 6**, Database storage options

Module 1

Overview of SQL Server

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Module Overview

- The SQL Server platform
- SQL Server Architecture
- Services and configuration options

Lesson: The SQL Server platform

- Components of SQL Server
- Client tools
- Instances
- Editions
- Version history

Components of SQL Server

- Not only the database engine
- Relational and business intelligence/data analysis components

SQL Server Components	
Database Engine Full-text search Replication Machine learning services Data Quality Services Polybase	Analysis Services
Integration Services	Reporting Services
Machine learning services (R, Python, Java)	Master Data Services

Client tools

- All are separate downloads and installers
- SQL Server Management Studio (SSMS)
 - Typically used by the DBA
 - New major version with new version of SQL Server
 - SQL Server 2019 – SSMS 18
 - SQL Server 2022 – SSMS 19
 - A higher version of SSMS can connect to a lower version of SQL Server
 - Includes Azure Data Studio
- Azure Data Studio (ADS)
 - Focus on developing T-SQL scripts
 - Has Extensions
- SQL Server Data Tools (SSDT)
 - Develop the database, creating tables, stored procedures etc
 - Generates a DACPAC file that can be used for deployment

Download SQL Server Management Studio (SSMS)

<https://docs.microsoft.com/en-us/sql/ssms/download-sql-server-management-studio-ssms>

Download and install Azure Data Studio

<https://docs.microsoft.com/en-us/sql/azure-data-studio/download-azure-data-studio>

Download SQL Server Data Tools (SSDT) for Visual Studio

<https://docs.microsoft.com/en-us/sql/ssdt/download-sql-server-data-tools-ssdt>

Instances

- Many SQL Server components are instance-aware
- Can have more than one installed, of
 - Database engine
 - Analysis services
- Instances enable isolation of:
 - Administration and security configuration
 - Performance and SLAs
 - Versions and collations
- Two types of instance:
 - Default instance, can have only one
 - Named instance

Database Engine Instances (SQL Server)

<https://docs.microsoft.com/en-us/sql/database-engine/configure-windows/database-engine-instances-sql-server>

Editions

- Enterprise
- Standard
- Web
- Developer (free, not for production)
- Express (free)
- Microsoft's cloud
 - Azure SQL Database
 - Azure SQL Managed Instance

Editions and supported features of SQL Server

<https://docs.microsoft.com/en-us/sql/sql-server/editions-and-components-of-sql-server-version-15>

Version history

Release Name	Version Number	Release Year	Service packs?
1.0	1.0	1989	Y
1.1	1.1	1991	Y
4.2	4.2	1992	Y
4.21	4.21	1994	Y
6.0	6.0	1995	Y
6.5	6.5	1996	Y
7.0	7.0	1998	Y
2000	8.0	2000	Y
2005	9.0	2005	Y
2008	10.0	2009	Y
2008 R2	10.5	2010	Y
2012	11.0	2012	Y
2014	12.0	2014	Y
2016	13.0	2016	Y
2017	14.0	2017	N
2019	15.0	2019	N
2022	16.0	2022 ?	

Demo Identify version and edition

History of Microsoft SQL Server

https://en.wikipedia.org/wiki/History_of_Microsoft_SQL_Server

What's new in SQL Server

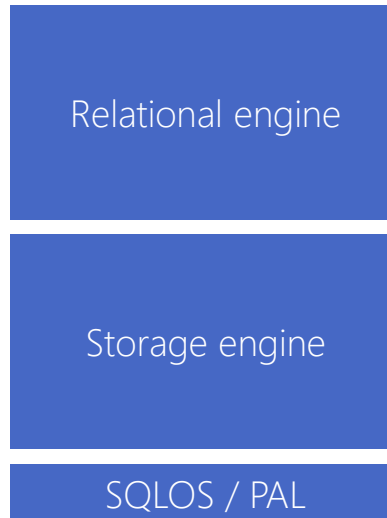
<https://karaszi.com/new-in-version-x>

Lesson: SQL Server Architecture

- SQL Server architecture
- How SQL Server uses CPU
- Parallelism
- Virtual Address Space
- Memory management
- Physical I/O and Logical I/O

SQL Server architecture

- Relational engine
 - Parses and optimizes the queries, and manages the caching and execution of query execution plans
- Storage engine
 - Manages buffer pages, I/O to the physical files, locking and transactions
- SQLOS layer
 - Provides an abstraction layer over common OS functions, providing task and memory management
 - Also known as PAL
 - Platform Abstraction Layer



SQL Server Architecture Explained: Named Pipes, Optimizer, Buffer Manager
<https://www.guru99.com/sql-server-architecture.html>

How SQL Server uses CPU

- Windows uses preemptive scheduling of threads
- One scheduler for every logical CPU created in SQL OS:
 - Manages the operating system threads and assigns tasks to threads
 - Minimizes context switches through cooperative scheduling
- CPU availability can be configured without restart:
 - A scheduler can be enabled or disabled
- Tasks waiting on a resource are moved to a waiting list:
 - Wait type and time are exposed through DMV
 - Details are useful for monitoring and troubleshooting

Understanding SQL Server Schedulers, Workers and Tasks

<https://www.mssqltips.com/sqlservertip/4403/understanding-sql-server-schedulers-workers-and-tasks/>

CPU Scheduling Basics – Windows and SQL Server

<https://www.nocentino.com/posts/2016-05-26-cpu-scheduling-basics-windows-and-sql-server/>

SQL SERVER : SQL Server Scheduler

<https://www.sqlservercentral.com/blogs/sql-server-sql-server-scheduler>

Parallelism

- Parallelism refers to multiple processors cooperating to execute a single query at the same time
- SQL Server can decide to distribute queries to more than one task
 - Tasks can run in parallel
 - Overall execution is probably faster
 - Synchronization overhead is incurred
 - Parallelism is only considered plans estimated over a certain cost
- **Max degree of parallelism**
 - How many CPUs can be used for execution of a parallel query operator
 - Can be overridden using database setting and MAXDOP query hint
- **Cost threshold for parallelism**
 - Minimal cost for considering parallel plans
 - You probably want to raise this from the default value of 5
 - Somewhere between 50 and 100 are commonly used values nowadays

Configure the max degree of parallelism Server Configuration Option

<https://docs.microsoft.com/en-us/sql/database-engine/configure-windows/configure-the-max-degree-of-parallelism-server-configuration-option>

Understanding and Using Parallelism in SQL Server

<https://www.red-gate.com/simple-talk/databases/sql-server/learn/understanding-and-using-parallelism-in-sql-server/>

Parallelism in SQL Server Execution Plan

<https://www.mssqltips.com/sqlservertip/5404/parallelism-in-sql-server-execution-plan/>

Configure the cost threshold for parallelism Server Configuration Option

<https://docs.microsoft.com/en-us/sql/database-engine/configure-windows/configure-the-cost-threshold-for-parallelism-server-configuration-option>

Virtual Address Space

- Virtual Address Space is the memory that can be allocated to application
 - Such as SQL Server
 - VAS is 128 TB as of Windows Server 2012
 - It was 8 TB in earlier 64 bit Windows versions
- SQL Server performance strongly depends on memory

Memory Management Architecture Guide

<https://docs.microsoft.com/en-us/sql/relational-databases/memory-management-architecture-guide>

Memory management

- Buffer pool is the main memory object of SQL Server:
 - Holds data cache
 - Provides memory for other SQL Server components
 - Is divided into 8 KB pages
- SQL OS automatically allocates as much memory as is needed:
 - Has a mechanism to prevent memory shortage on the system
 - Can be configured using min and max server memory options

Physical I/O and Logical I/O

I/O Type	Description
Physical I/O	Physical I/O occurs when the requested page is not available in buffer cache and must be read from the data file into the data cache before the requested page can be supplied or when a changed page is written to the data file
Logical I/O	Logical I/O occurs when the requested page is available in the data cache

Demo Server configuration

Lesson: Services and configuration options

- SQL Server services
- Service account options
- Communication architecture
- Configuring network protocols

SQL Server services

- A SQL Server instance is made up of a number of Windows services
- Only the services required to support the installed SQL Server features will be present
- Some services are installed once per SQL Server instance
 - If a service is linked to an instance, the instance name will appear in brackets after the service name
 - For instance for database engine:
 - Default instance :MSSQLSERVER
 - An instance named "A": MSSQL\$A
 - Applies to Database Engine and Analysis Services
- Use SQL Server Configuration Manager to configure services

SQL Server Services

<https://docs.microsoft.com/en-us/sql/tools/configuration-manager/sql-server-services>

SQL Server Services

<https://www.virtual-dba.com/blog/sql-server-services/>

Service account options

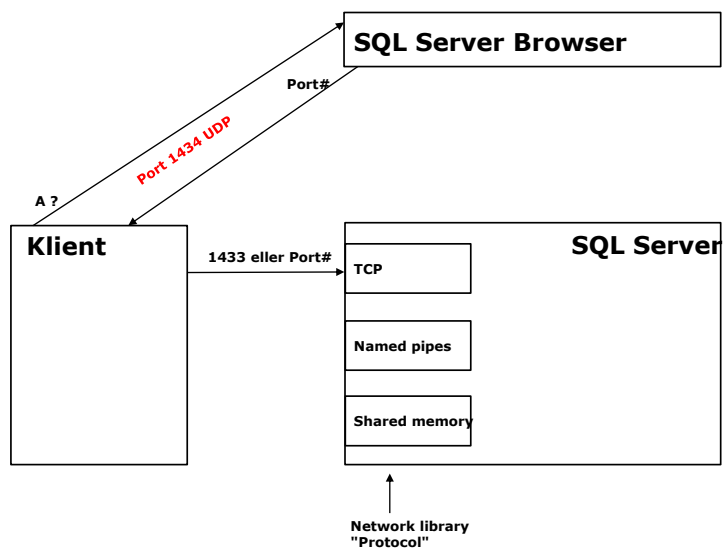
- Assign service accounts during installation
 - After installation, configure SQL Server services by using the SQL Server Configuration Manager tool
 - Do not change using the Services app
- Choose an appropriate account type:

• Managed service account	Domain	Good
• Domain user account	Domain	OK?
• Local user account	Local	Nah
• Local system account	Local	Absolutely not recommended
• Network service account	Local	Not recommended
• Virtual service account	Local	Good, default in setup

Configure Windows Service Accounts and Permissions

<https://docs.microsoft.com/en-us/sql/database-engine/configure-windows/configure-windows-service-accounts-and-permissions>

Communication architecture



Network Protocols and Network Libraries

<https://docs.microsoft.com/en-us/sql/sql-server/install/network-protocols-and-network-libraries>

Configuring network protocols

- Three network protocols are supported:
 - TCP/IP
 - Not enabled on Express and Developer edition by default
 - Named pipes
 - Not enabled by default
 - Shared Memory
- Network protocols can be defined for both server and client components
 - Server settings and client settings are configured independently
- You can use client aliases or DNC records to allow a server instance to respond to multiple names

SQL Server Configuration Manager

<https://docs.microsoft.com/en-us/sql/relational-databases/sql-server-configuration-manager>

No lab for this module

- Questions?

Module 2

Installing SQL Server

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Module Overview

- Planning and preparation
- Tempdb installation options
- Performing the installation
- Automating installation

Lesson: Planning and preparation

- Hardware and software requirements
- CPU and memory requirements
- Storage I/O and Performance
- Testing I/O correctness using SQLIOSIM
- Testing I/O performance using DISKSPD
- Virtual Servers and containers
- Installing SQL Server on Linux

Hardware and software requirements

- See the documentation for the version of SQL Server you are about to install
- Below are for SQL Server 2019
- Hardware requirements:
 - Processors 2019: 1.4 min, 2.0 rec
 - Memory 2019: 1 GB
 - Disk 2019: 6 GB available
- Software requirements:
 - Operating system 2019: WS 2016+, Win 10 TH1 1507 +
 - Prerequisite software .NET, network; both are in OS

SQL Server 2019: Hardware and software requirements

<https://docs.microsoft.com/en-us/sql/sql-server/install/hardware-and-software-requirements-for-installing-sql-server-ver15>

SQL Server 2016 and 2017: Hardware and software requirements

<https://docs.microsoft.com/en-us/sql/sql-server/install/hardware-and-software-requirements-for-installing-sql-server>

- Required resources are dependent on many factors
 - The database schema
 - Queries
 - Indexes
 - Load type (OLTP, analytic queries)
 - Size of data
 - Number of users
- You basically have three options
 - Test using a realistic workloads
 - Use recommendations from the vendor or developer
 - Give it a shot, and then adjust accordingly

Testing I/O correctness using SQLIOSim

- Used for integrity tests on disk subsystems
- Uses same I/O pattern as SQL Server
- Tests contain an element of randomness
- React if there are any errors (**red**) in the result log
- SQLIOSIM is **not** usable for performance testing

Demo

SQL Server Database Engine Input/Output requirements

<https://learn.microsoft.com/en-US/troubleshoot/sql/admin/database-engine-input-output-requirements>

Use the SQLIOSim utility to simulate SQL Server activity on a disk subsystem

<https://docs.microsoft.com/en-us/troubleshoot/sql/tools/sqliosim-utility-simulate-activity-disk-subsystem>

The SQL Server Sqliosim Utility

<https://www.red-gate.com/simple-talk/databases/sql-server/database-administration-sql-server/the-sql-server-sqliosim-utility/>

Testing I/O performance using DISKSPD

- Load generator for I/O subsystems
- Can be configured to mimic SQL Server I/O
 - Specify 8 or 64 KB I/O units
- Designed for use as a performance-tuning tool
- Replaces SQLIO
- CrystalDiskMark is a GUI on top of Diskspd

Demo Diskspd

Getting Started with Diskspd

<https://www.brentozar.com/archive/2015/09/getting-started-with-diskspd/>

DISKSPD on Github

<https://github.com/microsoft/diskspd>

Installing SQL Server on Linux

- Trends toward open-source systems
- Trends in virtualization
- SQL Server 2017 is the first version of SQL Server to run on Linux and containers

SQL Server on Linux

<https://docs.microsoft.com/en-us/sql/linux/sql-server-linux-overview>

Installation guidance for SQL Server on Linux

<https://docs.microsoft.com/en-us/sql/linux/sql-server-linux-setup>

Virtual Servers and containers

- Virtualization:
 - Virtual Machines
 - Hyper-V
- Containerization:
 - Containers
 - Docker
- SQL Server Installation on VMs and Containers
- Cloud Virtualization and Containerization

SQL Server Virtualization Performance Issues and Best Practices

<https://www.dnsstuff.com/sql-server-virtualization-performance-issues>

SQL Server VM Best Practices

<https://www.mssqltips.com/sqlservertip/6956/microsoft-sql-server-virtual-machine-best-practices/>

Support policy for Microsoft SQL Server products that are running in a hardware virtualization environment

<https://docs.microsoft.com/en-us/troubleshoot/sql/general/support-policy-hardware-virtualization-product>

Quickstart: Run SQL Server container images with Docker

<https://docs.microsoft.com/en-us/sql/linux/quickstart-install-connect-docker>

Deploy and connect to SQL Server Docker containers

<https://docs.microsoft.com/en-us/sql/linux/sql-server-linux-docker-container-deployment>

SQL Server on Docker

https://hub.docker.com/_/microsoft-mssql-server

Lesson: Tempdb installation options

- What is tempdb?
- Modifying the number of tempdb files during installation

What is tempdb?

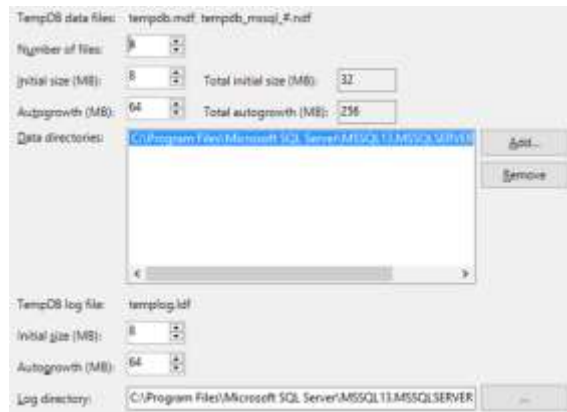
- **Tempdb** files are similar in architecture to data files for other SQL Server databases
- The usage pattern for **tempdb** is unlike other databases
 - Holds temporary objects created by users or by the database engine
- SQL Server recreates **tempdb** every time the instance restarts
- **tempdb** is a shared resource, so its performance can affect the performance of the whole system
- More about tempdb in a later module

tempdb database

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/tempdb-database>

Modifying the number of tempdb files during installation

- You can specify the following attributes of **tempdb** files during installation of SQL Server:
 - Location
 - Number
 - Initial size
 - Autogrowth increment



The screenshot shows the 'TempDB data files' configuration window in the SQL Server Enterprise Setup wizard. The window is titled 'TempDB data files: tempdb.mdf, tempdb_msdf_#.ndf'. It contains several input fields and buttons:

- Number of files:** A spin box set to 8.
- Initial size (MB):** A spin box set to 8. To its right, **Total initial size (MB):** is 32.
- Autogrowth (MB):** A spin box set to 64. To its right, **Total autogrowth (MB):** is 256.
- Data directories:** A list box containing the path 'C:\Program Files\Microsoft SQL Server\MSSQL11.MSSQLSERVER\'. To the right of the list box are 'Add...' and 'Remove' buttons.
- TempDB log file:** A text field containing 'templog.ldf'.
- Initial size (MB):** A spin box set to 8.
- Autogrowth (MB):** A spin box set to 64.
- Log directory:** A text field containing 'C:\Program Files\Microsoft SQL Server\MSSQL11.MSSQLSERVER\'. To its right is a 'Browse...' button.

Cheat Sheet: How to Configure TempDB for Microsoft SQL Server

<https://www.brentozar.com/archive/2016/01/cheat-sheet-how-to-configure-tempdb-for-microsoft-sql-server/>

Lesson: Performing the installation

- Options for installing SQL Server
- Performing an installation of SQL Server
- Setup 2022 installation dialogs
- Performing post-installation checks

Options for installing SQL Server

- Installation wizard
- Command prompt
- Configuration file

SQL Server installation guide

<https://docs.microsoft.com/en-us/sql/database-engine/install-windows/install-sql-server>

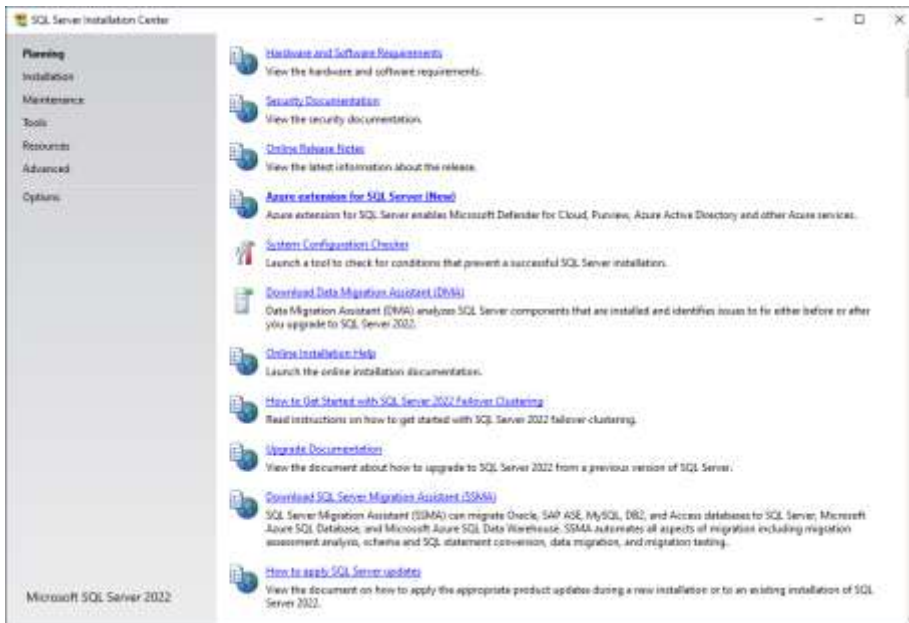
Performing an installation of SQL Server

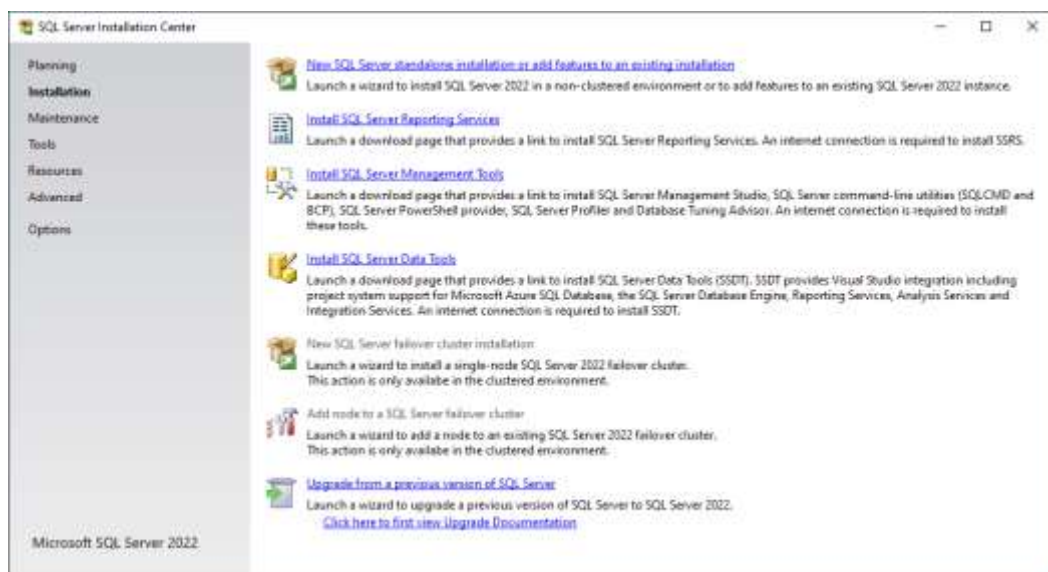
- Installation wizard:
 - Product Updates
 - Install Setup Files
 - Install Rules
 - Setup Role
 - Feature Selection
 - Instance Configuration
 - Server Configuration
 - <Component> Configuration
 - Ready to Install

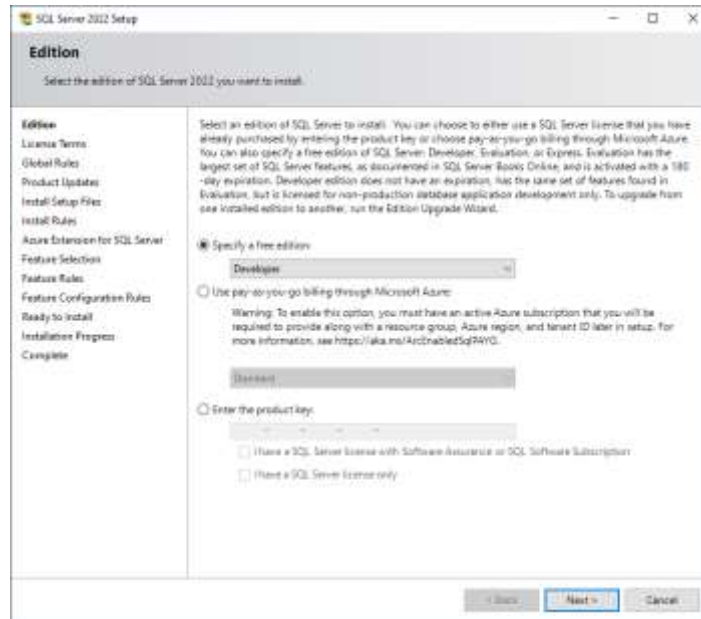
Installation Wizard help

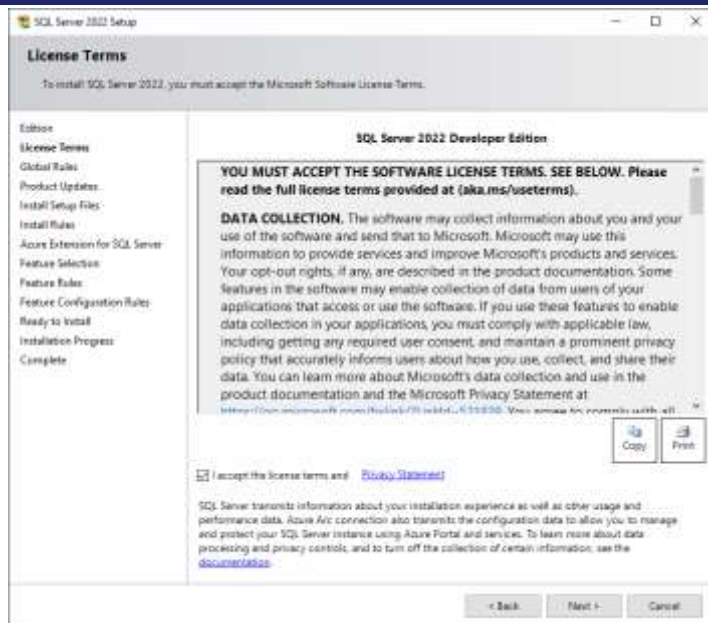
<https://docs.microsoft.com/en-us/sql/sql-server/install/instance-configuration>

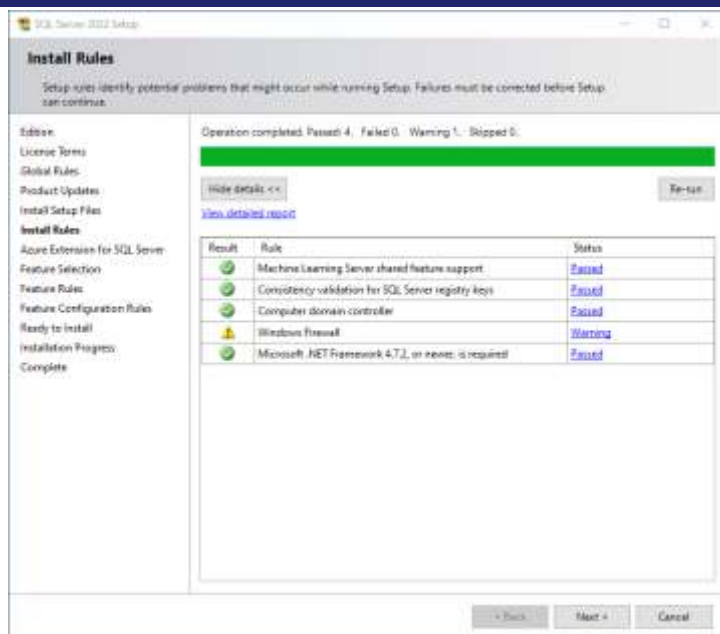
Setup 2022 installation dialogs











Introduced in 2022

SQL Server 2022 Setup

Azure Extension for SQL Server

Azure Extension for SQL Server is required to enable Microsoft Defender for Cloud, Purview, and Azure Active Directory.

☒ Azure Extension for SQL Server

To install Azure extension for SQL Server, provide your Azure account or a service principal to authenticate the SQL Server instance to Azure. You also need to provide the Subscription ID, Resource Group, Region, and Tenant ID where this instance will be registered. For more information for each parameter, visit: <https://aka.ms/ats-sql-server>.

☐ Use Azure Login

☒ Use Service Principal

Azure Service Principal ID*

Azure Service Principal Secret*

Azure Subscription ID*

Azure Resource Group*

Azure Region*

Azure Tenant ID*

Proxy Server URL (optional)

< Back Next > Cancel

SQL Server 2022 Setup

Feature Selection

Select the Developer features to install.

Edition

License Terms

Global Rules

Product Updates

Install Setup Files

Install Rules

Azure Extension for SQL Server

Feature Selection

Feature Rules

Instance Configuration

Server Configuration

Database Engine Configuration

Feature Configuration Rules

Ready to Install

Installation Progress

Complete

Looking for Reporting Services? [Download it from the web](#)

Features:

Instance Features:

- ☒ Database Engine Services
- ☒ SQL Server Replication
- ☐ Machine Learning Services and Language Ext
- ☒ Full-Text and Semantic Extractions for Search
- ☐ Data Quality Services
- ☐ PolyBase Query Service for External Data
- ☐ Analysis Services

Shared Features:

- ☐ Data Quality Client
- ☒ Integration Services
- ☐ Scale Out Master
- ☐ Scale Out Worker
- ☐ Master Data Services

Redistributable Features

Feature description:

Includes the Database Engine, the core service for storing, processing and securing data. The Database Engine provides controlled access and rapid transaction processing and also provides rich support for sustaining high

Prerequisites for selected features:

Already installed:

- Windows PowerShell 5.0 or higher
- Microsoft Visual Studio Tools for Application

To be installed from media:

Disk Space Requirements

Drive C: 1594 MB required, 622943 MB available

Select All Unselect All

Instance root directory: C:\Program Files\Microsoft SQL Server\

Shared feature directory: C:\Program Files\Microsoft SQL Server\

Shared feature directory (x64): C:\Program Files (x86)\Microsoft SQL Server\

< Back Next > Cancel

SQL Server 2022 Setup

Instance Configuration

Specify the name and instance ID for the instance of SQL Server. Instance ID becomes part of the installation path.

Edition

- License Terms
- Global Rules
- Product Updates
- Install Setup Files
- Install Rules
- Azure Extension for SQL Server
- Feature Selection
- Feature Rules
- Instance Configuration**
- Server Configuration
- Database Engine Configuration
- Feature Configuration Rules
- Ready to Install
- Installation Progress
- Complete

☒ Default instance

☐ Named instance: * MSSQLSERVER

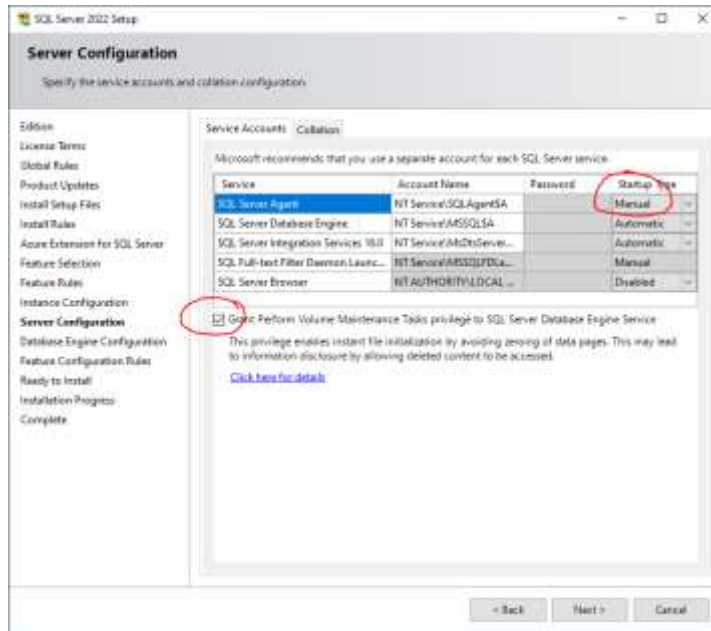
Instance ID: MSSQLSERVER

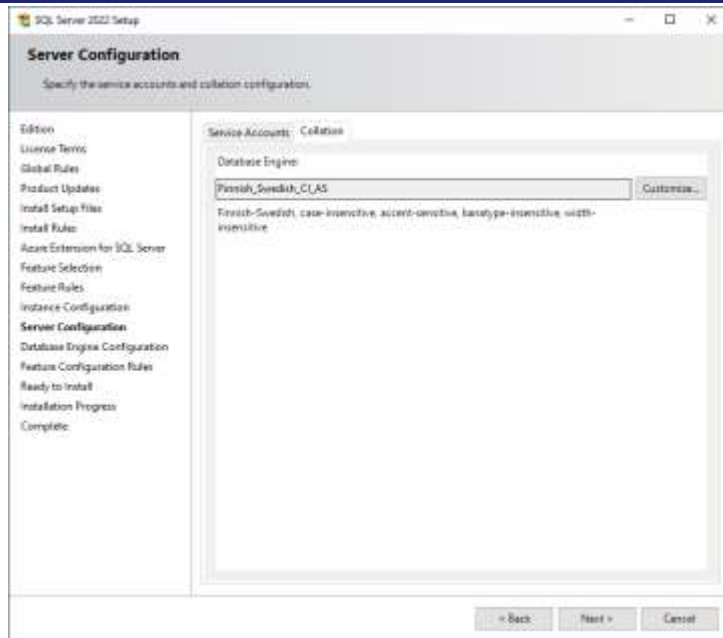
SQL Server directory: C:\Program Files\Microsoft SQL Server\MSSQL\SERVER

Installed instances:

Instance Name	Instance ID	Features	Edition	Version
---------------	-------------	----------	---------	---------

< Back Next > Cancel





SQL Server 2022 Setup

Database Engine Configuration

Specify Database Engine authentication security mode, administrators, data directories, TempDB, Max degree of parallelism, Memory limits, and FileStream settings.

Edition
License Terms
Global Rules
Product Updates
Install Setup Files
Install Rules
Azure Extension for SQL Server
Feature Selection
Feature Rules
Instance Configuration
Server Configuration
Database Engine Configuration
Feature Configuration Rules
Ready to Install
Installation Progress
Complete

Server Configuration | Data Directories | TempDB | MaxDOP | Memory | FILESTREAM

Specify the authentication mode and administrators for the Database Engine.

Authentication Mode

☐ Windows authentication mode

☒ Mixed Mode (SQL Server authentication and Windows authentication)

Specify the password for the SQL Server system administrator (sa) account.

Enter password:

Confirm password:

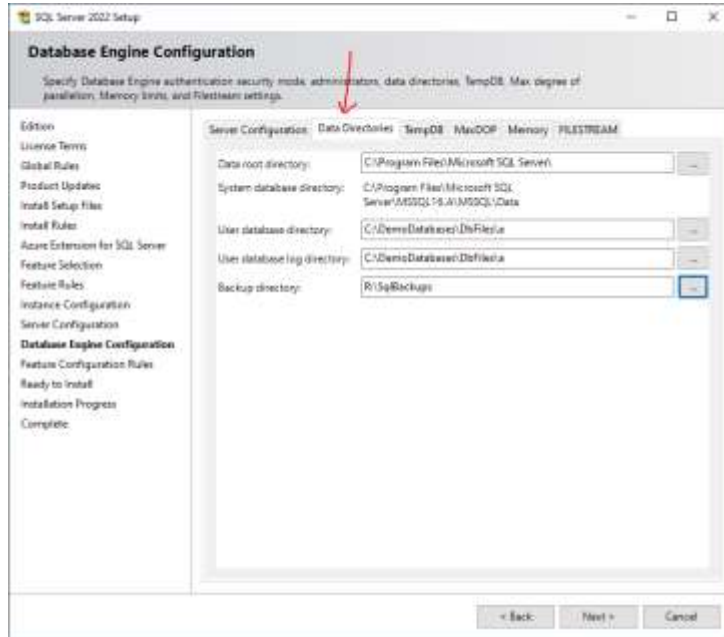
Specify SQL Server administrators

SQL Server administrators
sa

SQL Server administrators have unrestricted access to the Database Engine.

Add Current User Add Remove

< Back Next > Cancel



Introduced in 2016

SQL Server 2022 Setup

Database Engine Configuration

Specify Database Engine authentication security, mode, administrators, data directories, TempDB, Max degree of parallelism, Memory limits, and Filestream settings.

Edoos
License Terms
Global Rules
Product Updates
Install Setup Files
Install Rules
Azure Extension for SQL Server
Feature Selection
Feature Rules
Instance Configuration
Server Configuration
Database Engine Configuration
Feature Configuration Rules
Ready to Install
Installation Progress
Complete

Server Configuration Data Directories TempDB MaxDOP Memory FILESTREAM

TempDB data files: tempdb.mdf, tempdb.ndf

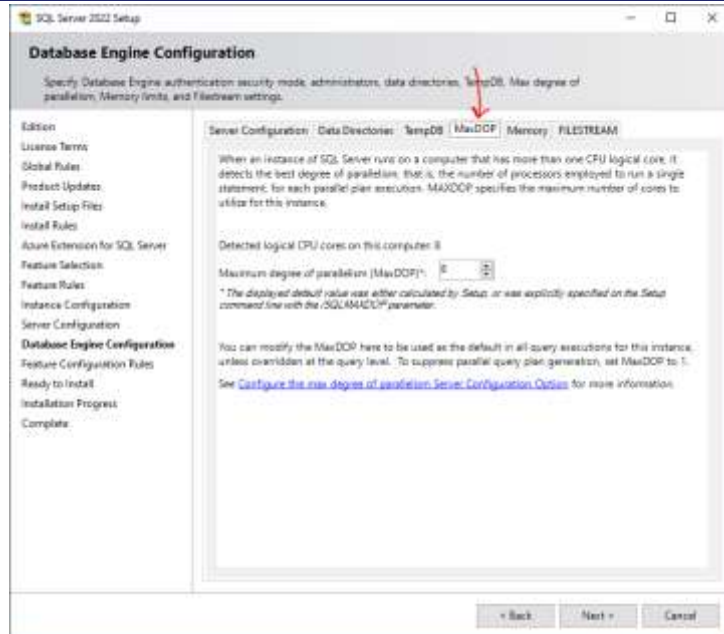
Number of files: 2
Initial size (MB): 8 Total initial size (MB): 16
Autogrowth (MB): 64 Total autogrowth (MB): 128

Data directories: C:\ProgramData\Microsoft SQL Server\TempDB\

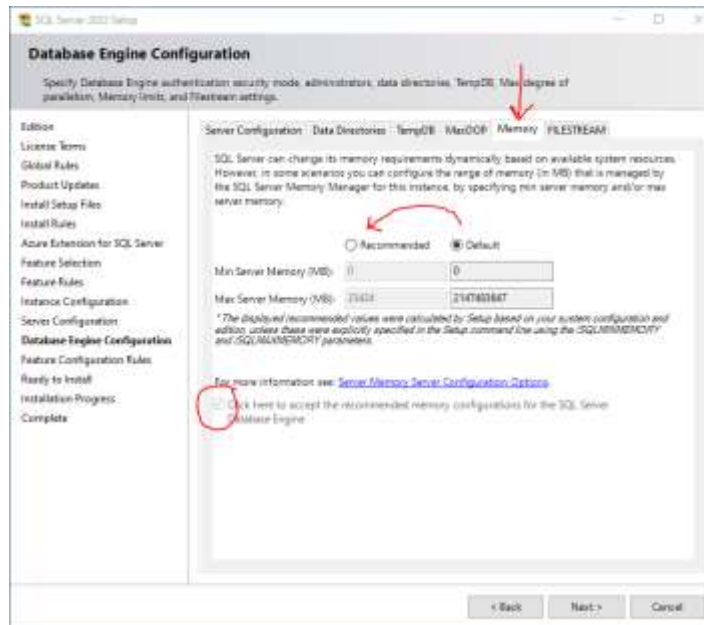
TempDB log file: templog.ldf
Initial size (MB): 8 Setup could take longer with large initial size.
Autogrowth (MB): 64
Log directory: C:\ProgramData\Microsoft SQL Server\TempDB\Log\

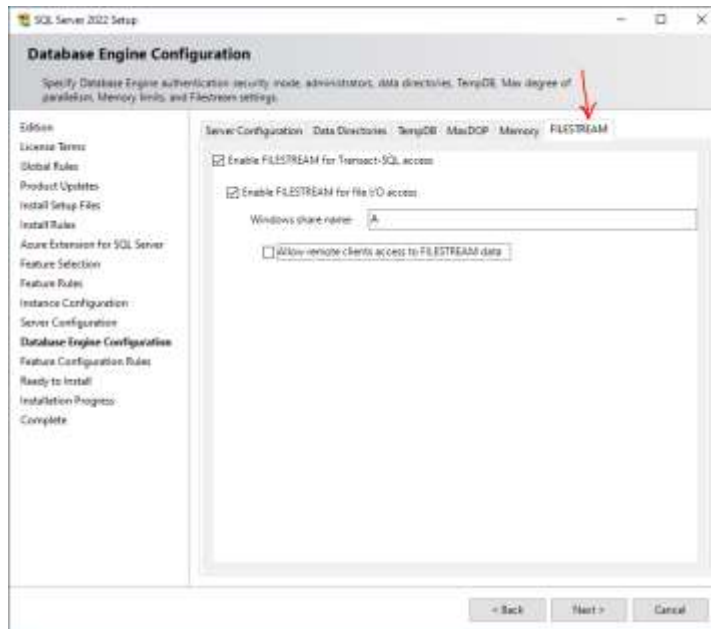
< Back Next > Cancel

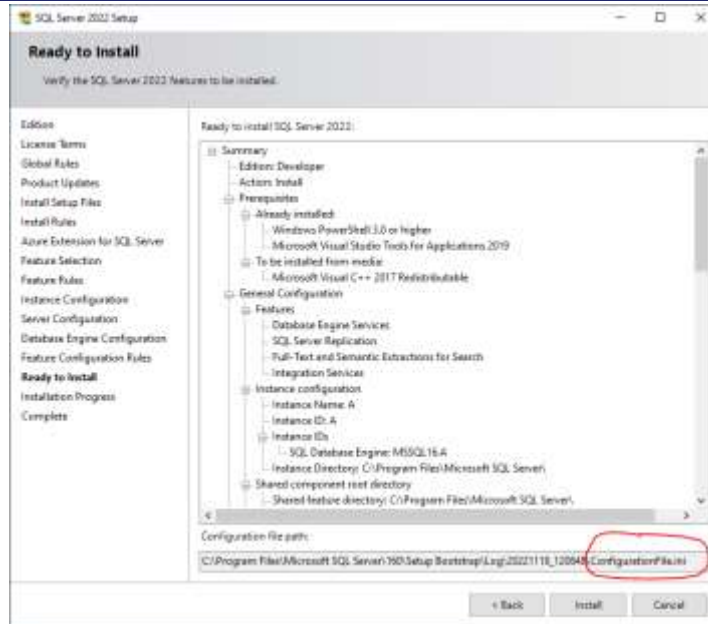
Introduced in 2019



Introduced in 2019







Performing post-installation checks

- Verify that SQL Server services are running
- If necessary, view log file information at: %ProgramFiles%\Microsoft SQL Server\150\Setup Bootstrap\Log

Lesson: Automating installation

- Unattended installation
- SQL Server Servicing

Unattended installation

- Install SQL Server from the command line:

```
Setup.exe /q /ACTION=CompleteImage /INSTANCENAME=MYNEWINST  
/INSTANCEID=<MYINST> /SQLSVCACCOUNT="<DomainName\UserName>"  
/SQLSVCPASSWORD="<StrongPassword>"  
/SQLSYSADMINACCOUNTS="<DomainName\UserName>"  
/AGTSVCACCOUNT="NT AUTHORITY\Network Service"  
/IACCEPTSQLSERVERLICENSETERMS
```

- Upgrade SQL Server from the command line:

```
Setup.exe /q /ACTION=upgrade /INSTANCENAME=MSSQLSERVER  
/RSUPGRADEDATABASEACCOUNT="<Provide a SQL Server logon account  
that can connect to the report server during upgrade>"  
/RSUPGRADEPASSWORD="<Provide a password for the report server upgrade  
account>" /ISSVCAccount="NT Authority\Network Service"  
/IACCEPTSQLSERVERLICENSETERMS
```

Demo ConfigurationFile.ini

Install SQL Server from the Command Prompt

<https://docs.microsoft.com/en-us/sql/database-engine/install-windows/install-sql-server-from-the-command-prompt>

Install SQL Server using a configuration file

<https://docs.microsoft.com/en-us/sql/database-engine/install-windows/install-sql-server-using-a-configuration-file>

SQL Server servicing

- Hotfixes
 - Fixes one issue
 - Not always public
 - Not extensively tested
- Security fix
 - Fixes one security issue
- Cumulative Updates (CU)
 - Rollup of several
 - Hotfixes
 - (Minor) improvements
 - Extensively tested
 - Service Packs
 - No longer, as of SQL Server 2017
 - Rollup of CUs
- GDR
 - Like CU but only security fixes
- Apply to test, UAT, etc before going to production

Install SQL Server Servicing Updates

<https://docs.microsoft.com/en-us/sql/database-engine/install-windows/install-sql-server-servicing-updates>

Why Should You Patch SQL Server?

<https://glennsqlperformance.com/2021/12/12/why-should-you-patch-sql-server/>

Lab 2: Installing SQL Server

- Ex 1. Before the installation
- Ex 2. Install a named instance
- Ex 3. Check the installation

Estimated Time: 45 minutes

Module 3

Upgrading SQL Server

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Module Overview

- Planning for upgrade
- Upgrade of the SQL Server components
- Side-by-side upgrade

Lesson: Planning for upgrade

- Creating an Upgrade Plan
- Supported Versions and Editions
- In-Place and Side-by-Side Upgrades
- Data Migration Assistant

Creating an Upgrade Plan

- Assess system to be upgraded
 - Which SQL Server features are installed?
 - Which features of SQL Server will be used after upgrade
 - Are all features in use supported by the new version of SQL Server?
 - What tools can assist?
 - Performance characteristics of the existing hardware?
- Confirm that upgrade is supported
- Select an upgrade method:
 - In-place
 - Side-by-side
- Check hardware resources
- Identify acceptance criteria
- Consider having Query Store enabled before the upgrade
 - Allows you to compare performance of queries before and after upgrade

Upgrade SQL Server

<https://learn.microsoft.com/en-us/sql/database-engine/install-windows/upgrade-sql-server>

Monitoring performance by using the Query Store

<https://docs.microsoft.com/en-us/sql/relational-databases/performance/monitoring-performance-by-using-the-query-store>

In-place or Side-by-side upgrade?

In-place	Side-by-side
Mostly automated	More granular control over process
System databases included	Useful for both test and production
No additional hardware	Can leverage failover/switchover
No change in server/instance name	
Rollback by reverting the whole machine Work since upgrade is lost!	Rollback by going back to old machine Work since upgrade is lost!

- Side-by-side upgrades can be carried out using one server or two servers
 - Using different instances
- You cannot restore or attach from a higher to a lower version

Choose a database engine upgrade method

<https://docs.microsoft.com/en-us/sql/database-engine/install-windows/choose-a-database-engine-upgrade-method>

Another reason I prefer side-by-side SQL Server instance upgrades

<https://sqlstudies.com/2021/06/29/another-reason-i-prefer-side-by-side-sql-server-instance-upgrades/>

In-place: supported versions and editions

- In-place upgrades to 2017 are supported from:
 - 2008 SP4 or later
 - 2008 R2 SP3 or later
 - 2012 SP2 or later
 - 2014
 - 2016
- In-place upgrades to 2019 are supported from:
 - SQL Server 2012 SP4 or later
 - SQL Server 2014 SP2 or later
 - SQL Server 2016 SP2 or later
 - SQL Server 2017
- In-place upgrades to 2022 will probably be supported from:
 - SQL Server 2012 SP4 or later
 - SQL Server 2014 SP2 or later
 - SQL Server 2016 SP2 or later
 - SQL Server 2017
 - SQL Server 2019
- Older versions require an interim upgrade
- In-place upgrades must take place between compatible editions

Supported version & edition upgrades (SQL Server 2019)

<https://docs.microsoft.com/en-us/sql/database-engine/install-windows/supported-version-and-edition-upgrades-version-15>

Supported version and edition upgrades (SQL Server 2022 Preview)

<https://learn.microsoft.com/en-us/sql/database-engine/install-windows/supported-version-and-edition-upgrades-2022?view=sql-server-ver16>

Data Migration Assistant

- Can assist in identifying breaking changes or deprecated features that might affect your application
- Can do the migration
 - SQL Server -> SQL Server uses backup and restore
 - Option to include Logins; will keep SIDs and passwords
- Download from Microsoft, not included on the installation media
- Can assess compatibility of T-SQL statements generated by client applications using:
 - Trace file containing queries
 - JSON files, prepared from C# client source code

Overview of Data Migration Assistant

<https://docs.microsoft.com/en-us/sql/dma/dma-overview>

Microsoft® Data Migration Assistant v5.6

<https://www.microsoft.com/en-us/download/details.aspx?id=53595>

Lesson: Upgrade of SQL Server Services

- Upgrading Analysis Services
- Upgrading Database Engine
- Upgrading Data Quality Services
- Upgrading Integration Services
- Upgrading Master Data Services
- Upgrading Power Pivot for SharePoint
- Upgrading Replicated Databases
- Upgrading and Migrating Reporting Services
- Upgrading SQL Server Management Tools
- Upgrading Using Setup

Upgrading Analysis Services (SSAS)

- In-place upgrade supported
- Side-by-side upgrade supported
 - Target SSAS instance must be configured with the same server mode (Tabular or Multidimensional) as the source SSAS instance

Upgrade Analysis Services

<https://docs.microsoft.com/en-us/sql/database-engine/install-windows/upgrade-analysis-services>

Upgrading Database Engine

- In-place upgrade supported
- Side-by-side upgrade supported
 - Objects stored in system databases must be migrated to the target instance, including:
 - Logins
 - SQL Server Agent jobs
 - Server-level triggers
 - etc

Upgrade Database Engine

<https://docs.microsoft.com/en-us/sql/database-engine/install-windows/upgrade-database-engine>

Moving a database between two SQL Server instances

<https://karaszi.com/moving-a-database-between-two-sql-server-instances>

Upgrading Data Quality Services

- In-place upgrade supported
- Side-by-side upgrade supported
- DQS database is treated like any other user database
- DQS schema must be upgraded as a separate step
 - DSQLINSTALLER.EXE
- All Data Quality Client instances must be upgraded using the setup utility

Upgrade Data Quality Services

<https://docs.microsoft.com/en-us/sql/database-engine/install-windows/upgrade-data-quality-services>

Upgrading Integration Services

- In-place upgrade not supported
- Side-by-side upgrade supported
- Behaviour of the upgrade will vary when SSIS and the database engine are installed on the same machine and upgraded together
 - Same machine:
 - New tables for SSIS packages -> new version of client is needed
 - Only database engine: No changes made to the tables
- Upgrade will not update SSIS packages to the new format
 - Wizard to upgrade packages (SSMS, SSDT)

Upgrading Master Data Services

- In-place upgrade supported
- Side-by-side upgrade supported
- MDS database schema must be updated
- MDS web application must be installed
- MDS Add-In for Excel must be updated
- You can in-place upgrade MDS without upgrading the database engine

Upgrade Master Data Services

<https://docs.microsoft.com/en-us/sql/database-engine/install-windows/upgrade-master-data-services>

Upgrading Replicated Databases

- In-place upgrade supported
- Side-by-side upgrade supported
- Nodes may run different versions of SQL Server
- Distributor version must be greater than or equal to the Publisher version
- Transactional replication subscribers must be within two versions of the Publisher

Upgrade or patch replicated databases

<https://docs.microsoft.com/en-us/sql/database-engine/install-windows/upgrade-replicated-databases>

Upgrading and Migrating Reporting Services

- In-place upgrade might be supported
 - From lower to 2016: Yes
 - From lower to 2017: No
 - From 2017 or higher to higher: Yes
- Side-by-side upgrade supported
 - One-server side-by-side upgrade only supported in native mode

Upgrade and Migrate Reporting Services

<https://docs.microsoft.com/en-us/sql/reporting-services/install-windows/upgrade-and-migrate-reporting-services>

Upgrading SQL Server Management Tools

- In-place upgrade not supported
- Side-by-side upgrade supported
- PATH environment variable should be checked to confirm which version of the tools will be accessed by default

In-place: upgrading using setup

- SQL Server setup program has wizards for:
 - In-place upgrade
 - Changing the edition of an existing SQL instance
- Upgrade can also be performed from the command line using setup.exe

Lesson: Side-by-side Upgrade: Migrating SQL Server Data and Applications

- Database version and compatibility level
- Using Backup and Restore to Migrate Databases
- Using Detach and Attach to Migrate Databases
- Using the Copy Database Wizard to Migrate Databases
- Transferring Logins and Passwords
- Orphaned Users
- Update Configuration with `sp_configure`

Database version and compatibility level

- Database file metadata is tightly linked to the version of SQL Server running the database
 - File metadata is automatically updated during a restore or attach operation
 - It is only possible to upgrade file metadata from an older version of SQL Server to a newer version
 - Rollback planning needs to take account of this
- Database compatibility level provides partial backward compatibility with previous versions of the database engine
- Watch out for performance differences when changing database compatibility level
 - Manage this using the Query Store functionality
 - Flight-recorder of queries, execution plans and run-time statistics

Demo Compatibility level

Compatibility Certification

<https://docs.microsoft.com/en-us/sql/database-engine/install-windows/compatibility-certification>

Using Backup and Restore to Migrate Databases

- Suitable for upgrading databases from SQL Server 2008 or later
- Pros:
 - Backup only includes used extents, not free space
 - Free space is not backed up
 - Backups can be compressed
 - Incremental restores of transaction log backups can be used to minimize downtime
 - Source database remains unchanged useful for
 - Rollback, go back to the old SQL Server
 - Testing, you can do this as many times as you want
- Cons:
 - Disk space required for backup file(s) and database file(s)
 - Downtime equals backup time *plus* transfer time *plus* restore time
 - Source database remains available—manage client application access

Copy Databases with Backup and Restore

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/copy-databases-with-backup-and-restore>

Using Detach and Attach to Migrate Databases

- Suitable for upgrading databases from SQL Server 2008 or later
- Pros:
 - Detach/attach commands are typically quick
 - Detaching the source database makes it unavailable—a clear cut-off point for client application changes
 - Copying the data files enables easy rollback (attach the files back to source server)
- Cons:
 - Large file sizes—database files typically include free space within the file
 - Moving data files (rather than copying) is a one-way process and cannot be rolled back—restore from backup would be required
 - Downtime equals detach time *plus* transfer time *plus* attach time
- More about detach/attach in next module

Detach a database

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/detach-a-database>

Using the Copy Database Wizard to Migrate Databases

- Detach/Attach provides a UI for detach/attach commands
- SQL Server Management Objects (SMO) method (script)
- Pros:
 - Can copy server-level objects (logins, SQL Agent jobs)
 - Source database remains unchanged and available
- Cons:
 - Slower than backup/restore or detach/attach
 - Vulnerable to network problems
 - Client application activity must be managed
- More about Copy Database Wizard in next module

Use the Copy Database Wizard

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/use-the-copy-database-wizard>

Transferring Logins and Passwords

- SQL Server logins are not included in a database backup or database files
 - Logins must be created on the target SQL Server when migrating databases
- SQL Server logins require a password to be specified:
 - Specify the existing password (if known)
 - Use the source system password hash to create the new login
- Contained databases with contained authentication uses no logins
 - Authorization is done against the user directly
 - No logins to transfer/re-map

Transfer logins and passwords between instances of SQL Server

<https://docs.microsoft.com/en-us/troubleshoot/sql/security/transfer-logins-passwords-between-instances>

Orphaned Users

- Occurs when a database user has no corresponding login, typically when:
 - The login was deleted
 - The database has been restored/attached from another SQL Server instance
- Use `sp_change_users_login` to report on orphaned users
- To fix orphaned users:
 - Use `sp_change_users_login` to fix users with SQL Server logins
 - `ALTER USER WITH LOGIN` to fix users with SQL Server or Windows logins

Demo Orphaned users

Moving a database between two SQL Server instances

<https://karaszi.com/moving-a-database-between-two-sql-server-instances>

Update Configuration with sp_configure

- sp_configure offers a convenient way to compare instance configuration settings between SQL Server instances
- **show advanced options** must be enabled for all settings to be displayed
- The list of settings might vary between versions of SQL Server
- sp_configure can also be used to change setting values

sp_configure (Transact-SQL)

<https://docs.microsoft.com/en-us/sql/relational-databases/system-stored-procedures/sp-configure-transact-sql>

Server Configuration Options (SQL Server)

<https://docs.microsoft.com/en-us/sql/database-engine/configure-windows/server-configuration-options-sql-server>

Lab 3: Upgrading SQL Server

- Ex 1. Create login and user on the source server
- Ex 2. Create and restore a backup
- Ex 3. Manage an orphaned user
- Ex 4. Raise the compatibility level

Estimated Time: 45 minutes

Module 4

Managing database storage and configuration options

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Module Overview

- Physical database structure
- Managing storage for system databases
- Managing storage for user databases
- Moving and copying database files and databases

Lesson: Physical database structure

- The physical storage of a database
- Considerations for Disk Storage Devices
- Determining File Placement and Number of Files
- Ensuring Sufficient File Capacity

How Data Is Stored in SQL Server



Example of database with

- Two file groups
- Three data files
- Two transaction log files

Database Files and Filegroups

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/database-files-and-filegroups>

Physical architecture of SQL server

<https://dbacentrals.blogspot.com/2017/07/physical-architecture-of-sql-server.html>

How Data Is Stored in SQL Server

Pages and extents



.ldf **.ldf**

0	1	2	3	4	5	6	7
8 Blue	9 Blue	10 Blue	11 Green	12 Green	13 Blue	14 Red	15 Green
16 Blue	17 Red	18 Blue	19 Blue	20 Green	21	22	23
24 Blue	25 Blue	26	27	28	29	30	31
32	33	...					

Pages and Extents Architecture Guide

<https://docs.microsoft.com/en-us/sql/relational-databases/pages-and-extents-architecture-guide>

Considerations for disk storage

- Direct Attached Storage: disks connected by a RAID controller
- Storage Area Network: disks connected by a network and available to multiple servers
- Windows Storage Pools: commodity disk drives grouped together to create one large storage space
- More about storage in later module

Determining file placement and number of files

- If applicable and single disks are used:
 - Isolate log and data files at the physical disk level
 - Not applicable when using SANs
 - Many disks are at the physical level
- Determine the number and location of data files based on performance and maintenance considerations
 - Use additional files to spread data across storage locations
 - Use smaller data files when easier maintenance is needed
 - You can backup and restore at the data file level
 - This is a more advanced backup scenario
- Determine log file requirements
 - Use a single log file in most situations - log files are written sequentially

Database Files and Filegroups, Recommendations

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/database-files-and-filegroups#Recommendations>

Ensuring Sufficient File Capacity

- Estimate the size of data, log files and tempdb:
 - Perform load testing with the actual application
 - Check with the application vendor
- Set the size to a reasonable size:
 - Leave enough space for new data, without the need to regularly expand
 - Monitor data and log file usage
 - Plan for manual expansion
 - Keep autogrowth enabled to allow for unexpected growth

Estimate the Size of a Table

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/estimate-the-size-of-a-table>

Estimate the Size of a Database

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/estimate-the-size-of-a-database>

TempDB Sizing Question

<https://stevestedman.com/2020/06/tempdb-sizing-question/>

SQL Server Transaction Log Architecture and Management Guide

<https://docs.microsoft.com/en-us/sql/relational-databases/sql-server-transaction-log-architecture-and-management-guide>

Manage the size of the transaction log file

<https://docs.microsoft.com/en-us/sql/relational-databases/logs/manage-the-size-of-the-transaction-log-file>

No but really, how big should my log file be?

<https://www.brentozar.com/archive/2016/02/no-but-really-how-big-should-my-log-file-be/>

Lesson: Managing storage for system databases

- SQL Server System Databases
- Moving System Databases
- Considerations for tempdb

SQL Server System Databases

System Database	Description
master	Stores various system-level configuration
msdb	Mainly used for SQL Server Agent
model	Template for new databases
tempdb	Holds various type of temporary data
resource	Source code for system procedures, views etc

System Databases

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/system-databases>

Moving System Databases

- Start with asking yourself:
 - “What problem am I solving by moving this system database?”
- Moving **msdb** and **model**, and **tempdb**
 - Execute ALTER DATABASE ... MODIFY FILE for each file
 - Stop the SQL Server service
 - Move the files
 - Restart the SQL Server service
- Moving **master**
 - Change the **–d** and **–l** startup parameters for the SQL Server service
 - Stop the SQL Server service.
 - Move the files
 - Restart the SQL Server service
- Misconfiguration can prevent SQL Server from starting

Move system databases

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/move-system-databases>

Considerations for tempdb

- **tempdb**:
 - Contains temporary data for internal objects, row versioning, and user objects
 - Is truncated/rebuilt with restart of the instance
 - Occupies varying amounts of space
 - Should be tested with real-life workloads
- Place **tempdb** on a fast (and separate?) I/O subsystem to ensure good performance
- Split **tempdb** into data files of equal size per core

Demo Tempdb

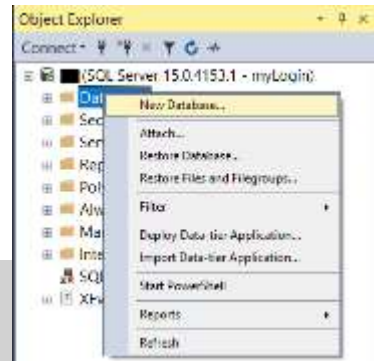
Lesson: Managing storage for user databases

- Creating User Databases
- Configuring Database Options
- Altering User Databases
- Managing Database Files
- Introduction to Filegroups
- Creating and Managing Filegroups

Creating user databases

- Create databases:
 - In SQL Server Management Studio
 - By using the CREATE DATABASE statement
 - Consider Instant File Initialization

```
CREATE DATABASE OlfData1
ON
( NAME = OlfData1,
  FILENAME = 'C:\DemoDatabases\DbFiles\OlfData1.mdf',
  SIZE = 20MB, MAXSIZE = 500MB, FILEGROWTH = 5MB ),
LOG ON
( NAME = OlfLog1,
  FILENAME = 'C:\DemoDatabases\DbFiles\OlfLog1.ldf',
  SIZE = 10MB, MAXSIZE = UNLIMITED, FILEGROWTH = 5MB )
```



Demo Create database

Create a Database

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/create-a-database>

CREATE DATABASE

<https://docs.microsoft.com/en-us/sql/t-sql/statements/create-database-transact-sql>

Configuring database options

- Database-level options are unique to each database

Option	Description
Auto options	Defines whether some operations should occur automatically within the database, for instance auto update statistics
Page verify	Defines how the page should be verified when read from disk; should be set to CHECKSUM
Recovery model	Defines the recovery model of the database
State options	Sets the state of the database, such as Online/Offline, Restricted Access or Read Only

Demo Database options

Database Properties (Options Page)

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/database-properties-options-page>

ALTER DATABASE (Transact-SQL)

<https://docs.microsoft.com/en-us/sql/t-sql/statements/alter-database-transact-sql>

ALTER DATABASE SET options (Transact-SQL)

<https://docs.microsoft.com/en-us/sql/t-sql/statements/alter-database-transact-sql-set-options>

ALTER DATABASE SCOPED CONFIGURATION (Transact-SQL)

<https://docs.microsoft.com/en-us/sql/t-sql/statements/alter-database-scoped-configuration-transact-sql>

Altering User Databases

- Altering a SET database option:

```
ALTER DATABASE Olf SET OFFLINE
```

- Altering database compatibility:

```
ALTER DATABASE Olf SET COMPATIBILITY_LEVEL = 150
```

- Altering a database scoped configuration:

```
ALTER DATABASE SCOPED CONFIGURATION SET MAXDOP = 8
```

Managing Database Files

- Adding space to a database:
 - ALTER DATABASE ... ADD FILE
 - ALTER DATABASE ... MAXSIZE
- Dropping database files:
 - Empty file: DBCC SHRINKFILE ... EMPTYFILE
 - Drop file: ALTER DATABASE
- Shrinking databases:
 - DBCC SHRINKDATABASE
 - DBCC SHRINKFILE
 - TRUNCATE_ONLY

Demo Alter database

Add Data or Log Files to a Database

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/add-data-or-log-files-to-a-database>

ALTER DATABASE (Transact-SQL) File and Filegroup Options

<https://docs.microsoft.com/en-us/sql/t-sql/statements/alter-database-transact-sql-file-and-filegroup-options>

Shrink a File

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/shrink-a-file>

DBCC SHRINKFILE (Transact-SQL)

<https://docs.microsoft.com/en-us/sql/t-sql/database-console-commands/dbcc-shrinkfile-transact-sql>

Why you want to be restrictive with shrink of database files

<https://karaszi.com/why-you-want-to-be-restrictive-with-shrink-of-database-files>

Introduction to Filegroups



Example of database with

- Two file groups
- Three data files
- Two transaction log files

Database Files and Filegroups

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/database-files-and-filegroups>

Creating and Managing Filegroups

- Creating filegroups
 - CREATE DATABASE ... FILEGROUP (<files>)
 - ALTER DATABASE ... ADD FILEGROUP <filegroup>
- Setting the default filegroup
 - ALTER DATABASE ... MODIFY FILEGROUP <filegroup> DEFAULT
- Using read-only filegroups
 - ALTER DATABASE ... MODIFY FILEGROUP <filegroup> READONLY

Demo File groups

Lesson: Moving and copying database files and databases

- Moving database files for user databases
- Detaching and Attaching Databases
- Use the Copy Database Wizard

Moving database files for user databases

- Data and log files can be moved within the instance:
 - Database must be offline
- ALTER DATABASE statement:
 - For copying within an instance
 - Manually move files on the file system

Demo Moving database files

Move User Databases

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/move-user-databases>

Detaching and Attaching Databases

- Moves or copies the database at the physical level
 - You get a binary copy of the database
 - Same goes for BACKUP and RESTORE
 - It either works or not
 - No in-betweens
- Detaching a database removes the database from the instance, but:
 - Data and log files are kept intact
 - Detached files can be attached again on the same or a different instance
- Use detach/attach to move databases to other instances
 - Detach/attach can be useful in disaster recovery situations

Demo Detach attach

Move a database using detach and attach (Transact-SQL)

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/move-a-database-using-detach-and-attach-transact-sql>

Detach a database

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/detach-a-database>

Attach a Database

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/attach-a-database>

Use the Copy Database Wizard

- Two options
 - A Wizard on top of detach/attach
 - Script and executes CREATE commands; and export/import the data
- How to use it
 - Select the source and destination servers
 - Choose the databases to move or copy
 - Specify where you want the databases located
 - Create logins for the destination server
 - Copy other database objects, jobs, user-defined stored procedures, and error messages
 - Schedule the Agent job on destination server that does the move or copy operation
- Can also used for side-by-side upgrade

Demo

Use the Copy Database Wizard

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/use-the-copy-database-wizard>

Other methods to do export / import of a database

- Generate a BACPAC file
 - A zip file containing
 - An XML file describing the database structure
 - Data files for your tables (BCP files)
 - Generate using SSMS, SQLPACKAGE.EXE or some other method
- Use "Generate Scripts" in SSMS
 - Right-click the database, Tasks, Generate Scripts...
 - Select Schema and data if you want both
 - This generates CREATE commands and INSERT commands

Demo

Use the Copy Database Wizard

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/use-the-copy-database-wizard>

Lab 4: Managing database storage

- Ex 1. Create a database
- Ex 2. Move tempdb
- Ex 3. Detach and attach a database

Estimated Time: 45 minutes

Module 5

Database maintenance

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Module Overview

- Database integrity
- Maintaining indexes and statistics
- Automating database maintenance

Lesson: Database integrity

- Introduction to database integrity
- Overview of DBCC CHECKDB
- Handling a data corruption situation
- DBCC CHECKDB options
- DBCC CHECKDB repair options

Introduction to database integrity

- Physical
 - Physical integrity
 - Data pages are written to physical storage as SQL Server requested and can be read correctly
 - Physical corruption
 - SQL Server does a request to OS and OS returns a failure (like a failed I/O request)
- Logical
 - Logical integrity
 - The data within the pages is logically correct so that they can refer to each other as required to instruct SQL Server to fetch related pages
 - Logical corruption
 - The I/O operation succeeded but the data wasn't what was written in the first place

Overview of DBCC CHECKDB

- Checks logical and physical database integrity
 - Allocation of all pages in the database
 - Consistency of tables and indexes
 - Consistency of the database catalog
 - ...and lots of other things
- Uses an internal database snapshot or table locks
- Should be run frequently
 - Synchronize executions with your backup strategy
 - RPO combined with backup strategy should decide the frequency for CHECKDB
- There are also commands such as
 - DBCC CHECKFILEGROUP
 - DBCC CHECKTABLE
 - ...

DBCC CHECKDB (Transact-SQL)

<https://docs.microsoft.com/en-us/sql/t-sql/database-console-commands/dbcc-checkdb-transact-sql>

CHECKDB From Every Angle: Complete description of all CHECKDB stages

<https://www.sqlskills.com/blogs/paul/checkdb-from-every-angle-complete-description-of-all-checkdb-stages/>

Paul S. Randal, In Recovery..., Category: CHECKDB From Every Angle

<https://www.sqlskills.com/blogs/paul/category/checkdb-from-every-angle/>

Handling a data corruption situation

- Be made aware that you have a data corruption
 - You have scheduled jobs to run DBCC CHECKDB
 - That also alerts you when they fail
- Stop production work and perform a tail log backup
- Restore the most recent healthy full (and possibly differential) backup
- Restore all subsequent log backups
- Determine why the corruption happened, "root cause analysis"
 - Else you might be waiting for a corruption to happen again
 - Fix it!
- Use Brent Ozar's blog post
 - In advance so you have an idea of what is coming
 - To hand-hold you through this process

What to Do When DBCC CHECKDB Reports Corruption

<https://www.brentozar.com/archive/2016/05/dbcc-checkdb-reports-corruption/>

Common bad advice around disaster recovery

<https://www.sqlskills.com/blogs/paul/common-bad-advice-around-disaster-recovery/>

DBCC CHECKDB Options

Option	Description
PHYSICAL_ONLY	Only checks the physical integrity to reduce overhead On by default in Maintenance Plans – <i>bad idea!</i>
NOINDEX	Does not perform logical checks on nonclustered indexes
ESTIMATEONLY	Estimates the amount of required tempdb space Uses a worst case scenario, a better estimate is to actually run it (possibly on dev/test)
TABLOCK	Uses locks instead of database snapshots
NO_INFOMSGS	Returns only error messages and no informational message
MAXDOP	Maximum number of CPU cores to use Parallel CHECKDB requires Enterprise Edition

Demo CHECKDB

DBCC CHECKDB (Transact-SQL)

<https://docs.microsoft.com/en-us/sql/t-sql/database-console-commands/dbcc-checkdb-transact-sql>

DBCC CHECKDB Repair Options

- Almost never a good idea!!!
- Database needs to be in SINGLE_USER mode
- DBCC CHECKDB output:
 - REPAIR_REBUILD
 - REPAIR_ALLOW_DATA_LOSS
- Instead of using repair, you should opt for:
 - Backup the last log, restore a healthy full backup, and all subsequent log backups
 - Possibly determine if the issue is in something that can be removed from the database, like an index

```
table error: object ID 149575571, index ID 1, partition ID 72057594059697408, alloc unit ID 720575940
CHECKDB found 0 allocation errors and 4 consistency errors in table 'Orders' (object ID 149575571).
CHECKDB found 0 allocation errors and 4 consistency errors in database 'CorruptDB'.
repair_allow_data_loss is the minimum repair level for the errors found by DBCC CHECKDB (CorruptDB).
```

Lesson: Maintaining indexes and statistics

- How indexes can be used
- Types of Indexes
- Dynamic Management Objects
- Index Fragmentation
- FILLFACTOR
- Ongoing Maintenance of Indexes
- Online Index Operations
- Updating Statistics

How indexes can be used

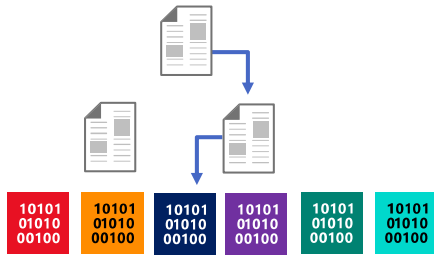
- Table Scan

- SQL Server reads all data pages



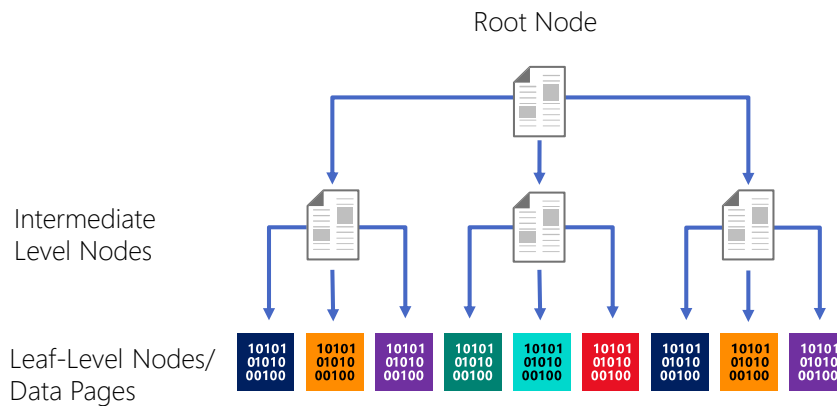
- Index

- SQL Server uses index pages to find rows



Types of Indexes

- Clustered index has data pages in the leaf level
- Nonclustered index has pointers to data rows in leaf level



Demo Simple index example

Clustered and nonclustered indexes described

<https://docs.microsoft.com/en-us/sql/relational-databases/indexes/clustered-and-nonclustered-indexes-described>

Dynamic Management Views and functions

- Frequently referred to as DMVs
 - Dynamic Management Views
 - Dynamic Management Functions
- Categories of DMVs
- Index-Related DMVs
 - sys.dm_db_index_usage_stats
 - sys.dm_db_index_operational_stats
 - sys.dm_db_index_physical_stats
 - sys.dm_db_missing_index_details

System Dynamic Management Views

<https://docs.microsoft.com/en-us/sql/relational-databases/system-dynamic-management-views/system-dynamic-management-views>

Index Related Dynamic Management Views and Functions (Transact-SQL)

<https://docs.microsoft.com/en-us/sql/relational-databases/system-dynamic-management-views/index-related-dynamic-management-views-and-functions-transact-sql>

Index Fragmentation

- Fragmentation occurs when data modification causes index pages to split:
 - Internal fragmentation: pages are not full
 - External fragmentation: pages are not in logical sequence
- It is likely that defragmenting index will not make a significant performance difference
 - We typically don't have single magnetic disks for our databases nowadays
- Detecting fragmentation:
 - Index properties in SQL Server Management Studio
 - `sys.dm_db_index_physical_stats`

Demo Index fragmentation and rebuild

Optimize index maintenance to improve query performance and reduce resource consumption
<https://docs.microsoft.com/en-us/sql/relational-databases/indexes/reorganize-and-rebuild-indexes>

Index fragmentation revisited
<https://sqlblog.karaszi.com/index-fragmentation-revisited/>

Ongoing Maintenance of Indexes

- Rebuild:

- Rebuilds the whole index
- Needs free space in database
- Performed as a single transaction

```
ALTER INDEX myIndex ON myTable REBUILD
```

- Reorganize:

- Sorts the pages and is always online
- Less transaction log usage
- Can be interrupted but still retain work performed to that point

```
ALTER INDEX myIndex ON myTable REORGANIZE;
```

SQLskills SQL101: REBUILD vs. REORGANIZE

<https://www.sqlskills.com/blogs/paul/sqlskills-sql101-rebuild-vs-reorganize/>

FILLFACTOR

- You can leave free space in indexes to avoid fragmentation:
 - FILLFACTOR
- Carefully consider what you expect to gain from doing this
 - Fill the pages to 100% and end up having page splits as soon as new data is inserted?
 - Or leave some free space on each page?
 - But that was the state before the defrag, so what did you gain from the rebuild in the first place?

```
ALTER INDEX myIndex ON myTable  
REBUILD WITH (FILLFACTOR = 80)
```

Online Index Operations

- Can create, rebuild, and drop indexes online:
 - Enables concurrent user access to the underlying table and indexes
 - Only needs short term shared locks at beginning and end of the operation and schema locks during the operation
- Typically slower than equivalent offline operation
- You can also specify max number of threads for the operation
- Requires Enterprise Edition

```
ALTER INDEX myIndex ON myTable  
REBUILD WITH (ONLINE = ON, MAXDOP = 4 )
```

Perform Index Operations Online

<https://docs.microsoft.com/en-us/sql/relational-databases/indexes/perform-index-operations-online>

Updating Statistics

- Distribution statistics become outdated
 - They can be set to update automatically/manually
 - Do not disable `auto_update_statistics`

Option	Description
AUTO_CREATE_STATISTICS AUTO_UPDATE_STATISTICS	Database options that enable SQL Server to automatically create/update statistics
UPDATE STATISTICS	Statement that updates all statistics on a table, or subset of statistics
sp_updatestats	System stored procedure that updates all statistics in a database

UPDATE STATISTICS (Transact-SQL)

<https://docs.microsoft.com/en-us/sql/t-sql/statements/update-statistics-transact-sql>

Make sure you update the statistics

<https://sqlblog.karaszi.com/make-sure-you-update-the-statistics/>

Stats Q&A 2: Updating SQL Server Statistics

<https://www.sqlskills.com/blogs/erin/update-statistics/>

Lesson: Automating database maintenance

- Regular database maintenance
- Database Maintenance Plans
- Ola Hallengren's maintenance solution

Regular database maintenance

- There are a number of things you might want to do at regular intervals:
 - Checking database integrity
 - Rebuilding or reorganizing indexes in a database
 - Probably don't result in noticeable improvements
 - But you pay for doing it:
 - Accessibility of the data while it is running
 - Generating all those log records
 - Updating statistics
 - Backup
 - Remove old history and backup files
- Some things are a really bad idea – don't do that:
 - Regular shrinks
 - Both reorganize and rebuild

Database Maintenance Plans

- GUI in SSMS to create a maintenance plan
- Creates:
 - An SSIS package, stored in msdb
 - One or more Agent jobs that execute the package
- Execution results stored in msdb and can also be:
 - Written to a text file
 - Sent to an operator
- Has somewhat of a bad reputation in the SQL Server community

Demo Maint plan

Maintenance Plans

<https://docs.microsoft.com/en-us/sql/relational-databases/maintenance-plans/maintenance-plans>

Use the Maintenance Plan Wizard

<https://docs.microsoft.com/en-us/sql/relational-databases/maintenance-plans/use-the-maintenance-plan-wizard>

Ola Hallengren's maintenance solution

- Widely used in the SQL Server community
- Available on Github
- More frequently downloaded from <https://ola.hallengren.com/>
- An installation script creates
 - A few stored procedures
 - A logging table (CommandLog)
 - A number of agent jobs
- It is your task to:
 - Configure the installation script, at the beginning
 - Where to write the backups, retention etc
 - Add a separate job for statistics update (the IndexOptimize job doesn't do that)
 - Modify the jobs
 - Schedule the relevant jobs
 - Email notification
 - ...

Demo Ola's scripts

SQL Server Maintenance Solution

<https://ola.hallengren.com/>

Ola Hallengren's SQL Server Maintenance Solution – Installation and SQL Server Backup solution

<https://www.sqlshack.com/ola-hallengrens-sql-server-maintenance-solution-installation-sql-server-backup-solution/>

Video: How to install Ola Hallengren "Maintenance Solution" TSQL

<https://www.youtube.com/watch?v=VxX5Z8vzO2Y>

Updating Your Ola Hallengren Scripts

<https://glennsqlperformance.com/2021/01/27/updating-your-ola-hallengren-scripts/>

Make sure you update the statistics

<https://sqlblog.karaszi.com/make-sure-you-update-the-statistics/>

Creating Schedules for Ola Hallengren's Maintenance Solution

<https://glennsqlperformance.com/2021/01/27/creating-schedules-for-ola-hallengrens-maintenance-solution/>

Lab 5: Database maintenance

- Ex 1. Run DBCC CHECKDB
- Ex 2. Create maintenance routines

Estimated Time: 45 minutes

Module 6

Database storage

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Module Overview

- Rarely used feature 1, SMB fileshare
- Rarely used feature 2, SQL Server storage in Microsoft Azure
- Rarely used feature 3, Storage Spaces in Windows Server
- Storage performance and recommendations

Lesson: Odd feature 1, SMB fileshare

- What Is SMB fileshare?
- SMB fileshare for database files
- Implementing SMB fileshare

What Is SMB fileshare?

SMB: Server Message Block

- Used for file and printer sharing
- Considerably improved in SMB 3.0
- Shares addressed using UNC paths
 - \\servername\share-name

Overview of file sharing using the SMB 3 protocol in Windows Server

<https://docs.microsoft.com/en-us/windows-server/storage/file-server/file-server-smb-overview>

Server Message Block

https://en.wikipedia.org/wiki/Server_Message_Block

SMB fileshare for database files

- It is supported to have database file on SMB
 - Since SQL Server 2012
- Potential benefits of SMB Storage for SQL Server
 - Cost
 - Manageability
- Disadvantage
 - Very rarely used

Install SQL Server with SMB fileshare storage

<https://docs.microsoft.com/en-us/sql/database-engine/install-windows/install-sql-server-with-smb-fileshare-as-a-storage-option>

Implementing SMB Fileshare

SMB Share Permissions:

- Full Control – SQL Server Service Account

Use UNC Path in Create Database statement

Share which cannot be used:

- Administrative Shares
- Loopback Shares
- Mapped Network Drives

Filestream on SMB is not supported

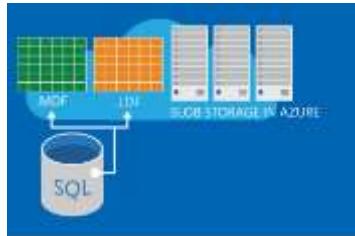
Demo SMB

Lesson: Odd feature 2, SQL Server storage in Microsoft Azure

- What are SQL Server Data Files in Azure?
- Benefits of SQL Server Data Files in Microsoft Azure
- Choosing a Storage Account and Performance Tier
- Implementing SQL Server Data Files in Azure

What are SQL Server Data Files in Azure?

- The database file are in Azure Blob Storage



SQL Server data files in Microsoft Azure

<https://docs.microsoft.com/en-us/sql/relational-databases/databases/sql-server-data-files-in-microsoft-azure>

Tutorial: Use Azure Blob Storage with SQL Server 2016

<https://docs.microsoft.com/en-us/sql/relational-databases/tutorial-use-azure-blob-storage-service-with-sql-server-2016>

Benefits of SQL Server Data Files in Microsoft Azure

- Potential benefits of SQL Server data files in Microsoft Azure
 - Easy and fast migration benefits
 - Cost and limitless storage benefits
 - High availability and disaster recovery benefits
 - Security benefits
 - Snapshot backups

Choosing a Storage Account and Performance Tier

- Storage Account Types:
 - General-Purpose Storage Accounts
 - Blob Storage Accounts
- Storage Account Performance Tiers:
 - Standard
 - Premium
- For SQL Server data files in Azure, create general-purpose storage accounts and use the standard tier

Implementing SQL Server Data Files in Azure

1. Create an Azure storage account
2. Add a container
3. Generate SAS key
4. Create a credential on the SQL Server instance
5. Create a database using the container credential

Lesson: Odd feature 3, Storage Spaces in Windows Server

- Storage Spaces
- Configuring Storage Pools and Storage Spaces

Storage Spaces

- Storage spaces

- Allow pooling of disks
 - Including internal and external IDE, SATA, SAS, USB
- Provide resiliency
- Flexible
 - Can be expanded easily

Storage Spaces overview

<https://docs.microsoft.com/en-us/windows-server/storage/storage-spaces/overview>

Configuring Storage Pools and Storage Spaces

1. Create a storage pool
2. Create a storage space
3. Create a volume

How to Combine Multiple Hard Drives With Windows' Storage Spaces

<https://www.pcmag.com/how-to/how-to-combine-multiple-hard-drives-with-windows-storage-spaces>

Lesson: SQL Server storage performance

- I/O patterns for data files (mdf and ndf)
- I/O patterns for transaction log files (ldf)
- Tempdb storage
- I/O performance
- I/O related wait statistics
- Recovery related aspects

I/O patterns for data files (mdf and ndf)

- Reads
 - Synchronous, when data is requested
 - Read-ahead (EE has deeper RA)
 - Buffer ramp-up (EE)
 - Merry-go-round scanning (EE)
 - One scan operation jumps in on some other on-going scan
 - Only for allocation order scans (IAM)
 - Requires heap and NOLOCK
- Writes
 - Asynchronous
 - Checkpoint
 - Lazywriter
 - Exception is minimally logged operations which are synchronous

Investigating I/O using Extended Events

<https://techcommunity.microsoft.com/t5/core-infrastructure-and-security/dissecting-sql-server-physical-reads-with-extended-events-and/ba-p/370393>

I/O patterns for transaction log files (ldf)

- Reads
 - Some special operations reads the transaction log
 - Rollback
 - Recovery
 - Start-up
 - Fail-over
 - Restore
 - Transactional replication
 - Change Data Capture
- Writes
 - Synchronous sequential writes

SQL Server Transaction Log Architecture and Management Guide

<https://docs.microsoft.com/en-us/sql/relational-databases/sql-server-transaction-log-architecture-and-management-guide>

- Already discussed...
- **tempdb**:
 - Contains temporary data for internal objects, row versioning, and user objects
 - Is truncated or rebuilt with every restart of the instance
 - Occupies varying amounts of space
 - Should be tested with real-life workloads
- Split **tempdb** into data files of equal size per core

I/O performance

- Is often critical to SQL Server performance
- Design the storage solution to meet your workload
- Use DMVs to assess the I/O workload for each file
 - `sys.dm_io_virtual_file_stats(dbid, fileid)`
- Glenn Berry's diagnostic queries has three queries for getting file stats at:
 - Drive level
 - Database level, compare to other databases
 - File level

```
SELECT *  
FROM sys.dm_io_virtual_file_stats(NULL, NULL)
```

`sys.dm_io_virtual_file_stats` (Transact-SQL)

<https://docs.microsoft.com/en-us/sql/relational-databases/system-dynamic-management-views/sys-dm-io-virtual-file-stats-transact-sql>

Troubleshoot slow SQL Server performance caused by I/O issues

<https://learn.microsoft.com/en-us/troubleshoot/sql/performance/troubleshoot-sql-io-performance>

How to examine IO subsystem latencies from within SQL Server

<https://www.sqlskills.com/blogs/paul/how-to-examine-io-subsystem-latencies-from-within-sql-server/>

What Virtual Filestats Do, and Do Not, Tell You About I/O Latency

<https://sqlperformance.com/2013/10/t-sql-queries/io-latency>

Glenn Berry's SQL Server Diagnostic Queries

<https://glennsqlperformance.com/resources/>

Linux Administration for SQL Server DBA's: Changing the I/O Scheduler

<https://www.mssqltips.com/sqlservertip/4844/linux-administration-for-sql-server-dbas-changing-the-io-scheduler/>

I/O related wait statistics

- Use DMVs to review wait statistics:
 - sys.dm_os_waiting_tasks
 - sys.dm_os_wait_stats
 - sys.dm_exec_session_wait_stats
- Wait types that relate to the storage system:
 - Page IO Latch waits
 - WRITELOG waits
 - LOGBUFFER waits
 - ASYNC_IO_COMPLETION waits
 - IO_COMPLETION waits
- Glenn Berry's diagnostic queries has a query that refines the data and also ignores irrelevant wait stats

sys.dm_os_wait_stats (Transact-SQL)

<https://docs.microsoft.com/en-us/sql/relational-databases/system-dynamic-management-views/sys-dm-os-wait-stats-transact-sql>

SQL Server Wait Types Library

<https://www.sqlskills.com/help/waits/>

Glenn Berry's SQL Server Diagnostic Queries

<https://glennsqlperformance.com/resources/>

Recovery related aspects

- Keep number of virtual log files to a reasonable number
 - Shortens recovery, failover etc.
- Consider turning on accelerated database recovery (2019)
 - Shortens recovery, failover etc
 - “Immediate” rollback
 - Less space used for transaction log
 - Further improvements in 2022
- Turn on indirect checkpoints
 - Default for new databases as of 2016
 - Lots of databases were born in an earlier version
 - Shorter checkpoints
 - Faster backups

SQL Server Transaction Log Architecture and Management Guide

<https://docs.microsoft.com/en-us/sql/relational-databases/sql-server-transaction-log-architecture-and-management-guide>

A Busy/Accidental DBA's Guide to Managing VLFs

<http://adventuresinsql.com/2009/12/a-busyaccidental-dbas-guide-to-managing-vlfs/>

Virtual Log Files

<https://dallasdbas.com/virtual-log-files/>

Accelerated Database Recovery in SQL Server 2019

<https://www.mssqltips.com/sqlservertip/5971/accelerated-database-recovery-in-sql-server-2019/>

ADR improvements in SQL Server 2022 (16.x) Preview

<https://docs.microsoft.com/en-us/sql/relational-databases/accelerated-database-recovery-concepts#adr-improvements-in->

More Reasons to Enable SQL Server Indirect Checkpoints

<https://www.mssqltips.com/sqlservertip/7108/sql-server-indirect-checkpoints-why-enable/>

No lab for this module

